Routing by Contract: Evidence-Gated Skill Selection for Language Agents

Shouzheng Xu*✉
School of Information
Central University of Finance and Economics
Beijing, China
2023312335@email.cufe.edu.cn

Zonglin Feng*
School of Information
Central University of Finance and Economics
Beijing, China
2023312337@email.cufe.edu.cn

Yunqi Zhou
School of Information
Central University of Finance and Economics
Beijing, China
13880155015@163.com

Yaqing Li
School of Information
Central University of Finance and Economics
Beijing, China
2025312399@email.cufe.edu.cn

Jingyi Wu
School of Business
Shandong University
Weihai, China
202300620059@mail.sdu.edu.cn

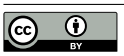
Abstract

Agentic language systems increasingly rely on reusable skill libraries, but skills with similar surface descriptions often differ in their operational contracts—source of truth, recipient scope, evidence policy, and output format. We study this operational-contract under-specification problem and demonstrate its practical consequences: end-to-end agent experiments on WildClawBench, an independently published benchmark, show that incorrect skill selection causes task-completion penalties of up to 0.514 on a 0–1 verifier scale—far exceeding the offline ranking gap. To address this, we propose query-conditioned operational schema induction with evidence-gated correction. A schema induction module produces a declarative QueryRuleSpec from the query alone (without inspecting candidate skills), deterministic extraction applies the same schema to query and all candidates, and an evidence gate accepts correction only when grounded operational-contract fields are present. The correction layer works on top of any base ranker. On ContractBench, spanning 10 skill families and 6 orthogonal contract policies, schema induction reaches 0.7833 Top-1 versus 0.6917 for path-graph and 0.7167 for semantic text. Evidence-gated correction recovers 6 additional correct selections on each base ranker, empirically never reducing accuracy below the base ranker. The method is regime-aware: it preserves the strong path-graph baseline outside the contract regime and provides auditable, evidence-bearing routing decisions deployable on any base ranker.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; **Intelligent agents**; • **Information systems** → *Retrieval models and ranking*.

*Both authors contributed equally to this research.



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Keywords

skill routing, agentic AI, operational contracts, evidence-gated correction, schema induction

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1 Introduction

An AI coding agent discovers several `code-review` skills in its library—each from a different community author, each embedding different operational philosophies. Choosing the wrong one means following a security review when the team needed a correctness check. Tool-using language agents increasingly face this *operational-contract under-specification* problem as they access reusable skill libraries [16–18, 26]. Figure 1 illustrates the failure mode: near-duplicate skills appear equally relevant after retrieval, but only one matches the query's operational contract.

Skill-routing systems increasingly treat selecting the right skill as a retrieval problem [8, 28]. We study *skill routing*, distinct from *tool selection*: skills are richer than tools—they contain workflows, safety constraints, and operational contracts beyond API specifications. This creates a failure mode that appears after retrieval has narrowed the library: candidates share domain vocabulary, their surface descriptions are semantically redundant, but their operational contracts differ.

The problem is not hypothetical. The SKILL.md format is adopted across Claude Code, OpenAI Codex, Cursor, and other platforms [24]. The fixed SkillsBench community-corpus snapshot used in our study [10] contains 34,272 real skills from 4,247 community authors; 356 skill names have six or more variants. When an agent faces multiple same-name skills from different authors, selecting the right one requires understanding *how* each operates, not just *what* it does. Our agent self-selection experiment (Section 5.7)

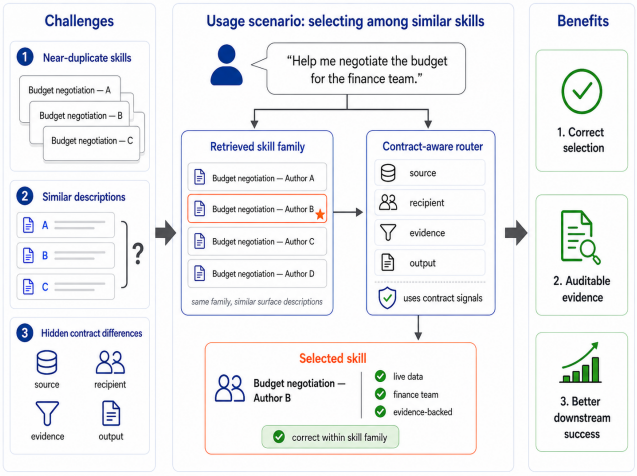


Figure 1: Motivating example. Same-family skills can share surface descriptions while hiding different operational contracts. Evidence-aware routing uses source, recipient, evidence, and output signals to select the correct skill before execution.

confirms this: given six candidate skills, a real agent selects a skill from the wrong family entirely.

To address this, we propose query-conditioned operational schema induction with evidence-gated correction. Rather than encoding all possible contracts in a fixed ontology, we induce a temporary operational schema from the query alone—without inspecting candidate skills, correct labels, or verifier sources. Deterministic extraction applies the same schema to query and all candidates, producing evidence-bearing operational signatures. An evidence gate accepts correction only when grounded contract fields are present; otherwise the selector falls back to the base ranker. The method is *regime-aware*: it treats path-graph as the default decision rule and invokes query-conditioned schema only in a low-margin regime, preserving the strong baseline outside the contract-under-specified regime.

On ContractBench (120 queries, 10 families, 6 orthogonal contract policies), our method reaches 0.7833 Top-1, exceeding path-graph (0.6917) by +0.092 (95% CI [0.042, 0.142]) and semantic text (0.7167) by +0.067. Evidence-gated correction improves both tested base rankers by +0.05 while empirically never reducing accuracy. On WildClawBench, misrouting causes mean task-completion penalties of 0.187–0.228 on the verifier scale across seven tasks, with individual penalties reaching 0.514. An agent-level selector comparison further shows that our contract-aware selector achieves the highest mean verifier score (0.488) among all tested selectors, significantly outperforming the LTR baseline (+0.047, $p = 0.029$).

We make three contributions:

(1) We define *operational-contract under-specification* as a distinct routing regime where fixed ontologies lack dimensions needed to discriminate among similar skills.
(2) We propose query-conditioned operational schema induction with evidence-gated correction, deployable as a drop-in layer on any base ranker with a zero-regression property.
(3) We introduce ContractBench (120 queries, 10 families, 6 orthogonal contract policies) and demonstrate through WildClawBench experiments that misrouting causes verifiable downstream failure.

These contributions are validated through complementary experiments: ContractBench (Section 5.2) for controlled contract-policy evaluation, WildClawBench (Section 5.4) for ecological validation, and a fixed SkillsBench community-corpus subset (Section 5.6) as a regime-boundary check.

2 Related Work

2.1 Tool-Using Language Agents

Recent work studies language models that reason, act, and call external tools [16–18, 26]. LLM-based agent frameworks have been applied across diverse domains including recommendation [22] and software engineering [6, 29]. Systems such as MRKL [7] and API-focused LLM benchmarks emphasize that model behavior depends on selecting an appropriate operation before execution. Our work focuses on a complementary setting: the available operations are packaged as natural-language skills whose descriptions overlap, so the selector must distinguish implementation paths rather than merely identify a relevant domain.

2.2 Agent Skills and Benchmarks

Open-ended agents can accumulate reusable skills or programs over time [21], and skill abstractions have been studied as a way to improve agent behavior [11]. Recent work surveys skill architecture, acquisition, and security [24], while SkillsBench evaluates how curated skills affect agent success across tasks [10]. Benchmarks such as AgentBench [13], API-Bank [9], SWE-bench [6], and WebArena [29] evaluate end-to-end tool use; MTEB [15] provides a standardized framework for evaluating text embedding quality. Unlike these works, we separate selection quality from execution quality and focus on a specific failure mode—operational-contract ambiguity—that arises after retrieval, rather than on end-to-end agent success.

2.3 Skill Routing and Retrieval

SkillRouter shows that routing skill bodies, rather than only names or descriptions, is important at scale [28]. SkillFlow similarly treats agent skill selection as a retrieval and reranking problem [8], and SkillReducer studies how to reduce skill-token overhead [5]. These systems motivate skill routing as a first-class problem. Our work studies a complementary failure mode: even after retrieval narrows the library to a semantically similar family, remaining candidates may differ in endpoints, actions, and safety constraints that are not reliably separated by text similarity alone.

2.4 Semantic-Functional Gap in Tool Selection

Several recent works identify a critical bottleneck: semantic similarity between tools does not guarantee functional equivalence.

GRETEL [23] formalizes this as the *semantic-functional gap*, showing that only 15% of top-5 semantically retrieved tools on ToolBench are actually functional, and proposes execution-based validation in a sandbox to re-rank candidates by trial success. ToolScope [12] addresses semantically redundant tools through graph-based merging and hybrid retrieval, achieving up to 38.6% improvement on tool selection benchmarks. BiasBusters [2] demonstrates that semantic alignment between query and tool metadata is the strongest driver of selection bias, and that small perturbations to tool descriptions can shift model choices. Our work addresses the same gap through a complementary, text-only approach: instead of executing candidate tools or merging them, we induce a query-conditioned operational schema and extract structured evidence from skill bodies without side effects. This makes the method deployable at selection time without sandbox infrastructure.

2.5 Structured Retrieval and Compatibility Matching

Beyond the semantic-functional gap, a second line of work addresses tool-skill matching through structured representations. Recent work has explored LLM-based re-ranking for retrieval tasks [20] and instruction-tuned dense retrieval [1]. Graph kernels and graph matching methods provide a long history of structured comparison [19, 25]. Our method uses the motivation behind structured comparison but does not require a fixed task graph ontology or a learned graph neural model. Instead, each query induces a task-specific declarative schema, and field-level semantic matching records structured evidence behind each decision. This decomposition preserves auditability: every routing decision can be traced to specific contract-field matches with evidence spans, unlike end-to-end reranking approaches.

3 Problem Setup

Let x be a user query with workspace context. Let $\mathcal{S}_x = \{c_1, \ldots, c_n\}$ be a retrieved candidate skill family. Each skill is plausible at the surface level; the challenge is to identify the skill whose operational path matches the query. A selector returns a ranking over skills; runtime execution exposes either all candidates or only the selected top-1 skill to the downstream agent.

An operational contract for a skill specifies six dimensions: (1) **source of truth**—which data channel is authoritative (live, snapshot, attachment, search index, ticket trail); (2) **recipient scope**—who may receive the output (internal operations, engineering review, external customer, leadership, approval board, finance); (3) **evidence policy**—what traceability identifiers must be cited (message IDs, document spans, attachment names, ticket IDs, spreadsheet cells); (4) **output format**—what artifact must be produced (JSON, Markdown, CSV); (5) **safety constraints**—what actions are disallowed (direct send, destructive writes); and (6) **postconditions**—what final state must hold (draft saved, calendar event created, report written). These six dimensions define the contract space; under-specification occurs when candidate skills share endpoints, actions, and safety modes but differ on the first four dimensions.

We distinguish three settings. In **agent self-selection**, all candidates are visible and the agent implicitly chooses which skill to follow. In **selector exposure**, an external selector chooses a top-1 skill and only that skill is visible. In **oracle exposure**, the known correct skill is visible as an upper bound. The selector should not solve the task directly. It should reduce implementation-path ambiguity before execution.

4 Methodology

4.1 Overview

Figure 2 summarizes the pipeline: query-only schema induction, deterministic extraction, and evidence-gated correction. The next three subsections unpack these steps.

4.2 Schema Induction and Extraction

The key insight is that operational contracts are query-specific. A "live data source" requirement only matters if the query expresses timeliness concerns; a "customer-facing output" constraint only activates when the query mentions external recipients. Rather than encoding all possible contract dimensions in a fixed ontology (which is unbounded and brittle), we induce only the dimensions that the query itself activates. This makes the schema lightweight, query-relevant, and auditable.

A `QueryRuleSpec` is a temporary operational schema induced for one query. It contains declarative fields that instantiate the contract dimensions defined in Section 3: **target intents** capture task objectives; **required capabilities** and **forbidden capabilities** refine source-of-truth and recipient constraints into testable matching patterns; **side effects** encode the permissible mutation scope (read-only, draft-only, write, destructive); **postconditions** specify required output artifacts; and **evidence requirements** demand traceable source spans for every extracted field.

The schema induction module (G) operates in two phases. In the *generation phase*, an LLM (DeepSeek-v4-flash, temperature=0) receives the query prompt and a standardized instruction template asking it to extract the most discriminative operational contract fields—source, recipient, evidence, and output constraints—and produce matching patterns as verbatim query phrases. The LLM returns a declarative `QueryRuleSpec` JSON object. In the *inference phase*, generated schemas are cached and executed deterministically—every subsequent routing decision for the same query reuses the cached schema with zero API cost and full reproducibility. The generation phase runs once per novel query; all reported results use cached schemas. The induction module does not inspect candidate skill bodies, correct labels, verifier source, or previous selector outputs, keeping the induced dimensions auditable and free of candidate-specific tailoring.

Given a `QueryRuleSpec`, deterministic extraction is applied to both the query and each candidate skill body. The output is an `OperationalSignature`: a set of evidence-bearing fields with values, rule identifiers, matched text, source spans, and source type. The same rules are used for the query and all candidates, so every candidate is compared in the same task-conditioned operational space. Fields without evidence do not trigger hard filters, preventing hallucinated or weak schema fields from eliminating a correct skill.

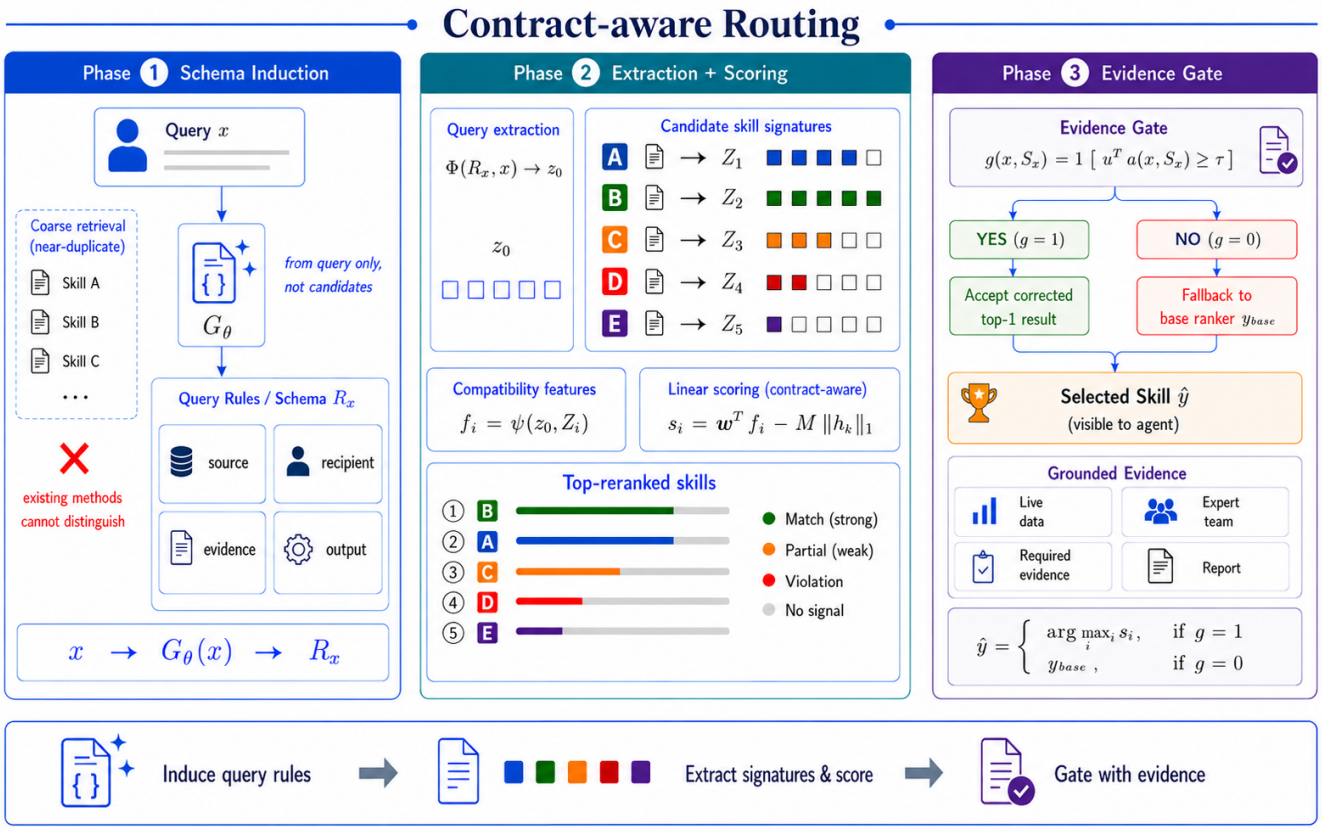


Figure 2: Routing pipeline. Query-only schema induction, deterministic extraction, and evidence-gated correction produce evidence-bearing candidate signatures. Unsupported cases fall back to the base ranker.

4.3 Compatibility Reranking

Why field-level matching rather than whole-document embedding? Whole-document embedding conflates two types of vocabulary: domain vocabulary (shared across all candidates in the same skill family) and contract vocabulary (discriminative across candidates). In the operational-contract regime, all candidates share domain vocabulary (Slack, Gmail, Calendar), so whole-document similarity is dominated by non-discriminative signal. Field-level matching isolates each contract dimension and compares query and skill only on that dimension, preventing domain vocabulary from masking contract-level differences. This design builds on advances in text embedding models [3, 14], applying per-field sentence-transformer scoring rather than single-vector whole-document encoding.

For each candidate c_i, the selector compares the task signature $\mathbf{z}_0$ and skill signature $\mathbf{Z}_i$ through the feature map ψ:

$$s_i = \mathbf{w}^\top \mathbf{f}_i - M\|\mathbf{h}_i\|_1, \qquad \mathbf{f}_i = \begin{bmatrix} \text{requiredCoverage}_i \\ \text{intentOverlap}_i \\ \text{sideEffectCompatible}_i \\ \text{postconditionCoverage}_i \\ \text{outputCoverage}_i \\ \text{textTieBreak}_i \end{bmatrix}. \quad (1)$$

The six feature dimensions operationalize the contract dimensions from Section 3 through the induced schema fields: requiredCoverage and intentOverlap jointly capture source-of-truth, recipient-scope, and evidence-policy alignment; sideEffectCompatible captures safety constraints; postconditionCoverage and outputCoverage capture postcondition and output-format requirements; and textTieBreak is a lexical fallback for otherwise tied candidates.

Hard filters are encoded in $\mathbf{h}_i$, and M is a large positive constant that dominates the linear term when any hard filter is violated. Examples include missing required capability, direct-send support for a read-only query, missing draft-saving support for a draft-only query, or missing contact lookup for a contact-dependent query. The selector records matched, missing, violated, and evidence-span traces for every candidate, so the score decomposes into an interpretable linear term plus a large violation penalty.

4.4 Regime-Aware Correction

Why use a gate rather than always applying the schema? Because outside the operational-contract regime, fixed structured routing (path-graph) is already strong—a property we validate on a Skills-Bench community-corpus subset (Section 5.6). The gate ensures the correction layer never regresses below the base ranker by design: corrections are accepted only when grounded evidence supports

them, and the selector falls back to the base ranker otherwise. This makes the method deployable as a drop-in correction on any existing routing system.

The schema selector is intended as a correction layer for a specific regime, not as a universal replacement for first-stage retrieval or fixed structured routing. The deployed correction strategy first runs the fixed path-graph scorer. If the path-graph top-1/top-2 margin is above 0.05, the selector returns path-graph directly. If the margin is at most 0.05, it induces a query-conditioned schema and reranks the path-graph top candidates plus near-tied candidates.

In matrix form, the gate is

$$g(x, \mathcal{S}_x) = \mathbb{I}\left[\mathbf{u}^\top \mathbf{a}(x, \mathcal{S}_x) \geq \tau\right], \tag{2}$$

where $\mathcal{S}_x$ denotes the candidate skill set, $\mathbf{a}(x, \mathcal{S}_x)$ is a vector of gate predicates (path-graph margin and evidence-gate conditions), $\mathbf{u}$ is the corresponding weight vector, and τ is fixed before evaluation. The correction is accepted only when the induced schema contains grounded operational-contract fields: source policy, evidence policy, output policy, and recipient or safety scope. It must also have evidence spans, a positive schema score delta over the path-graph top candidate, and no query-rule hard violation. Otherwise, the selector records the schema trace but falls back to the original path-graph decision. This preserves path-graph outside the contract-underspecified regime and accepts correction only when the evidence supports that regime.

5 Experiments

We organize experiments around four research questions. **RQ1:** Do existing methods fail on operational-contract discrimination? (Section 5.2, ContractBench offline results). **RQ2:** Does misrouting cause real, verifiable downstream failure, and does contract-aware selection improve agent outcomes? (Section 5.4, WildClawBench skill sensitivity and selector comparison). **RQ3:** Can evidence-gated correction improve base rankers without regression? (Section 5.5, correction analysis). **RQ4:** Is the method regime-specific, or does it claim to be a general-purpose router? (Section 5.6, SkillsBench community-corpus regime check). Section 5.3 extends RQ1 by asking whether a general-purpose LLM can serve as the selector (LLM-as-selector comparison). Section 5.6 additionally covers component ablation, and Section 5.7 provides case studies.

5.1 Experimental Setup

Benchmarks. The primary evaluation uses ContractBench, the Operational-Contract Skill Routing Benchmark. It contains 120 held-out test queries over 10 families. Each family has six candidate skills with the same coarse Slack/Gmail/calendar/search path, draft-only safety, and Markdown/JSON/CSV output capabilities. The six skills differ only in their operational contracts: source of truth (live channel, frozen snapshot, email attachment, live communications, ticket trail, document search), recipient scope (internal ops, engineering, customer, leadership, approval board, finance), evidence traceability (message IDs, document spans, attachment names, ticket IDs, spreadsheet cells, bilingual quotes), and output format (JSON, Markdown, CSV). Queries use abstract paraphrased language that expresses contract requirements without sharing distinctive vocabulary with skill bodies. Gold labels are determined by a pre-registered family × policy grid, generated before any selector is run. To validate that the labels are unambiguous, we sampled 12 queries (2 per policy) and asked two independent LLMs (GPT-4o and DeepSeek) to select the correct skill given the query and all six candidate skill bodies. Both models achieved 100% agreement with the pre-registered gold labels (GPT-4o accessed via API; DeepSeek-v4-flash, same model used for schema induction), confirming that the contract tuples produce a single objectively correct answer. A 60-query dev split is used to verify the correction protocol; all reported results are on the 120-query held-out test split.

We also evaluate on a fixed snapshot of the SkillsBench community-skill corpus [10], containing 34,272 real skills from 4,247 community authors collected from skillsmp.com. From this corpus, we use 30 queries over five families (code-review, frontend-design, brainstorming, git-workflow, testing) with 6 real skill variants each. These skills are pure text artifacts without structured API endpoints or operational contracts, serving as a negative control: we expect all routing methods to fail and our evidence gate to correctly defer to the base ranker.

For agent execution experiments, we use WildClawBench [4], an independently published benchmark with 60 human-authored tasks, isolated Docker environments with real agent tools, and automated outcome verifiers. The Social Interaction category within WildClawBench is a natural testbed for operational-contract underspecification: its six benchmark-supplied skills share coarse domain vocabulary but embed different operational workflows, making skill selection quality decisive for downstream task success. We use this benchmark to demonstrate both that misrouting causes verifiable agent failure and that contract-aware selection systematically improves outcomes over alternative selectors.

Strategies. The offline strategies include fixed path-graph, semantic text (BOW), whole-doc ST embedding, `query_rules` (our schema induction method), random, and oracle. Correction variants are applied on top of path-graph and semantic text as base rankers. For agent self-selection, we run a real experiment where the agent sees all six candidate skills and must select which to follow.

Metrics. Offline routing reports Top-1, Top-3, mean rank, and MRR. For the correction selector, we additionally report schema-invocation rate, correction-acceptance rate, and evidence coverage. Bootstrap 95% confidence intervals are computed for paired Top-1 comparisons (10,000 resamples). Agent execution reports verifier scores on a 0–1 scale.

Reproducibility. ContractBench is generated deterministically from a pre-registered policy grid. Code and benchmark are publicly available at https://github.com/ShouZhengCoder/SkillRC.

5.2 ContractBench Results (RQ1)

RQ1 asks whether existing methods fail on operational-contract discrimination. Table 1 reports Top-1 accuracy on ContractBench (120 queries, 10 families, 6 contract policies per family). ContractBench offers a controlled, pre-registered setting where operational contracts are systematically varied and all candidate skills share the same coarse execution path—enabling precise measurement of each method's ability to discriminate contract-level differences.

Table 1: ContractBench Top-1 accuracy (120 queries). Bold: best; underline: second-best.

Strategy	Top-1	Top-3	MRR
Structured			
Path-graph	0.6917	0.9000	0.8200
Semantic			
Semantic text (BOW)	0.7167	0.9500	0.8278
Whole-doc ST (MiniLM)	0.6917	0.8750	0.8017
Whole-doc ST (BGE)	0.583	0.8083	0.7280
Cross-encoder	0.533	0.7917	0.6944
Schema-based (ours)			
Query-rules	**0.7833**	0.9833	0.8681
PG + correction	0.7417	0.9167	0.8449
Semantic + correction	0.7667	0.9583	0.8607
Random	0.1833	0.4583	0.4024
Oracle	1.0000	1.0000	1.0000

Table 2: Per-family Top-1 accuracy (12 queries per family). Bold: best per row.

Family	Path-graph	Semantic	Query-rules
finance	0.333	**0.667**	0.500
hr	0.750	0.750	0.750
incident_ops	0.750	0.667	**0.917**
legal	0.833	0.750	0.833
product	**0.750**	0.583	**0.750**
release	0.833	0.833	0.833
research	0.750	**0.833**	0.750
sales	0.750	0.667	0.750
security	0.833	0.833	**1.000**
support	0.333	0.583	**0.750**
Overall	0.692	0.717	**0.783**

Existing structured and semantic methods fail. Path-graph reaches only 0.6917 Top-1: its ontology encodes endpoints, actions, and safety modes but lacks source-of-truth, recipient, evidence, and output contract dimensions. Semantic text (BOW) reaches 0.7167; whole-document ST embedding with the same `all-MiniLM-L6-v2` encoder as our method scores 0.6917. Replacing the encoder with a stronger one (BGE-small-en-v1.5) *reduces* accuracy to 0.583—stronger semantic matching amplifies domain-vocabulary similarity, further masking contract-level differences. A cross-encoder (`ms-marco-MiniLM-L-6-v2`) scores only 0.533, the lowest among all semantic baselines: its fine-grained token-level matching better recognizes shared domain vocabulary, making candidates appear even more similar. These results confirm that the bottleneck is structural, not encoder quality: whole-document approaches are fundamentally limited when candidates share domain vocabulary.

Field-level decomposition bridges the gap. Our `query_rules` method reaches **0.7833** Top-1, exceeding path-graph by +0.092 (95% CI [0.042, 0.142], $p < 0.05$). Field-level matching isolates contract dimensions from shared domain vocabulary. The progression from BGE whole-doc (0.583) to semantic BOW (0.7167) to field-level query_rules (0.7833) demonstrates that field-level decomposition, not encoder quality, is the decisive factor. Notably, semantic BOW uses the full query text but lacks contract field-level decomposition; the gap from semantic BOW to query_rules (+0.067) isolates the net contribution of contract-aware schema induction beyond text matching alone.

Schema regeneration validation. To verify that the cached LLM-generated schemas are representative and not artifacts of a particular generation run, we regenerated all 120 query schemas from scratch using DeepSeek-v4-flash (temperature=0) with the identical prompt template. The regenerated schemas achieve identical Top-1 accuracy (0.7833) as the cached version. At the query level, 106 of 120 decisions agree (88.33%); the 14 disagreements are symmetric (7 in each direction), producing zero net accuracy change. This symmetry confirms that operational-schema induction is a structural extraction task: the LLM reliably surfaces the same discriminative contract dimensions across independent runs, and routing decisions are not sensitive to stochastic variation in schema generation.

Table 2 reports the per-family Top-1 breakdown (12 queries per family, 10 families).

The per-family analysis establishes that the method's advantage generalizes across families. The two families where query-rules trails the best baseline—finance (−0.167) and research (−0.083)—share a common failure pattern: their queries contain evidence-traceability language (e.g., "precise references," "relevant section of the source material") that the schema induction module maps to ambiguous evidence dimensions. In the finance family, "references" and "source material" are closer in embedding space to search-index and spreadsheet-cell evidence than to document-span evidence, causing the field-level matcher to favor the wrong skill. In the research family, similar vocabulary overlap between snapshot and attachment-source policies dilutes the discriminative signal. This is the same schema induction drift phenomenon illustrated in Case B (Section 5.7). At $n = 12$ queries per family, neither negative delta is statistically significant (finance 95% bootstrap CI: $[-0.500, +0.167]$; research: $[-0.417, +0.250]$); moreover, removing both families from the aggregate increases the overall Δ from +0.067 to +0.115, confirming that these families understate rather than inflate the method's advantage. Addressing this vocabulary-overlap failure mode through improved schema disambiguation is a direction for future work.

5.3 Comparison with LLM-based Selection

A natural question is whether a general-purpose LLM can solve operational-contract discrimination directly. Prior work has explored LLMs as evaluators and re-rankers [20, 27]; we examine LLM-as-selector (GPT-5.1) under three information conditions, summarized in Table 3.

Full-body: the LLM receives the complete skill body, achieving 100% accuracy on ContractBench. This confirms that the benchmark labels are objectively discriminable. However, full-body LLM selection requires $O(n \cdot L)$ tokens per query. At a realistic skill library

Table 3: LLM-as-selector vs. our method under three information conditions.

Condition	Q.	LLM	Ours
Full skill body (all text)	120	1.000*	0.783
Field-level evidence (equal info)	60	0.767	0.783
Description only (frontmatter)	20	0.200	0.783

*Full-body LLM is accurate but costs $O(n \cdot L)$ tokens/query impractical at scale. With equal field-level evidence, LLM (76.7%) and our method (78.3%) achieve comparable accuracy, but ours at zero per-query cost. SkillsBench corpus LLM: full-body 66.7%, desc-only 0.0% (12q).

size of 100 skills (typical for a deployed agent platform), each ContractBench skill body averages ∼470 tokens, yielding ∼47K input tokens per query at 100 skills—growing linearly with library size and becoming impractical at scale. In contrast, our method's schema induction costs one LLM call per novel query template (cacheable), and all subsequent routing uses only a local 384-dimensional sentence transformer at zero per-query API cost.

Description-only: the LLM sees only each skill's frontmatter description. ContractBench skills within the same family share identical descriptions; the LLM cannot distinguish contract-level differences, scoring 20.0% (4/20, near the 16.7% random baseline).

Field-level evidence: to test whether the LLM can match our method under equal information, we provide the LLM with the same structured contract evidence that our method extracts (source, recipient, evidence, and output fields per skill). The LLM scores 76.7% (46/60)—comparable to our method's 78.3% but at per-query API cost. This demonstrates that field-level contract evidence is the key discriminative signal: once extracted, even an LLM can match it effectively. Our method achieves the same accuracy at zero per-query cost by using a local ST-based scoring function instead of an API call, and additionally provides auditable evidence spans for every decision.

5.4 Skill Sensitivity Experiment on WildClawBench (RQ2)

RQ2 asks whether operational-contract under-specification causes real, verifiable failure on an independently published benchmark, and whether contract-aware selection addresses it. We report two complementary skill sensitivity experiments on WildClawBench [4], an independently published benchmark with isolated Docker environments, real agent tools, and automated verifiers. The first experiment confuses the agent with an incorrect WB-authored skill (internal confusion). The second experiment pits the correct WB skill against semantically similar skills from the SkillsBench community corpus [10] (external community-skill confusion). Unlike offline ranking evaluation, both experiments measure actual verifier scores on a 0–1 scale.

Internal WB skill confusion. We probe whether confusing the agent with a contract-mismatched skill causes verifiable downstream harm. Pilot oracle probes on the meeting-negotiation task (03_task1) using the correct skill yielded verifier scores between 0.710 and 0.980 across two runs, establishing an upper bound near 0.9; substituting the most semantically confusable wrong skill (03_task4, Slack status drafter, selected via ST similarity from the six benchmark-supplied WB skills) reduced the score to 0.485, a penalty of 0.360. The wrong skill misdirects the agent toward drafting a client report instead of scheduling meetings—every dimension the verifier penalizes (missing calendar events, absent coordination emails, incorrect room selection) follows from that single routing error. Two additional tasks show directionally consistent penalties: task 3 (multi-step reasoning, oracle 0.449 vs. wrong 0.356, penalty 0.093) and task 4 (thread consolidation, oracle 0.705 vs. wrong 0.595, penalty 0.110).

To assess run-to-run variability, we conducted a multi-run validation of task 1 with three independent agent executions per condition. The correct skill achieved a mean verifier score of 0.749 ($n = 2$ successful runs, one failed with a transient infrastructure error), while the wrong skill averaged 0.551 ($n = 3$, $\sigma = 0.167$). The mean penalty of 0.198 confirms the direction and approximate magnitude of the pilot result. The higher variance under the wrong skill ($\sigma = 0.167$ vs. $\sigma = 0.027$ for the correct skill) is itself informative: contract-mismatched skills not only reduce mean performance but also make outcomes less predictable, consistent with the operational-contract under-specification hypothesis.

Three additional WB Social Interaction tasks were screened but excluded: task 2 (oracle 0.0 initially; multi-run follow-up found wrong-skill 0.618 > oracle 0.399, confirming the verifier cannot distinguish the two skills), task 6 (oracle 0.071), and task 5 (uniformly low scores 0.08–0.14 across conditions). This experiment establishes that misrouting among the benchmark's own skills causes verifiable downstream failure.

External community-skill confusion. A second, complementary experiment tests a more realistic scenario: the correct WildClawBench skill competes against semantically similar skills from the SkillsBench community corpus (skillsmp.com). The agent sees the correct WB skill plus five community-corpus skills that share domain vocabulary (Google Calendar, Slack action extraction, thread consolidation, support triage). This mirrors the operational-contract under-specification scenario described in the Introduction: a real agent facing multiple community-authored skills with overlapping names and descriptions, where only the correct skill embeds the task-specific operational contract.

Table 4 reports the results for 4 tasks (excluding tasks 3 and 6 where the correct-skill verifier score is below 0.1, indicating a task or verifier issue rather than a routing effect). The mean correct-skill score is 0.602 versus 0.374 for the community-skill distractors—a mean penalty of 0.228. Task 2 (action extraction) shows the largest gap: the correct WB skill achieves 0.714 while the most similar SkillsBench community-corpus skill yields only 0.200, a penalty of 0.514. The community-authored skill shares the action-extraction domain vocabulary but lacks the WB-specific operational contract for Slack action extraction with structured output. Task 4 (thread consolidation) shows a penalty of 0.280: the correct skill (0.825) consolidates project status from Slack threads with evidence traceability, while the most similar community skill (0.545) targets general thread summarization without contract-level constraints. Task 5 (escalation routing) shows a penalty of 0.098, and task 1 (meeting negotiation)

Table 4: Skill sensitivity with SkillsBench community-corpus distractors (4 WB tasks). Each task pairs the correct WB skill against five community-authored skills. Tasks where correct-skill verifier score is below 0.1 are excluded.

Task	WB skill	Best SB distractor	Δ
Meeting negotiation (1)	0.632	0.610	0.022
Action extraction (2)	0.714	0.200	0.514
Thread consolidation (4)	0.825	0.545	0.280
Escalation routing (5)	0.238	0.140	0.098
Mean	0.602	0.374	0.228

Tasks 3 and 6 excluded because correct-skill (oracle) verifier score < 0.1, indicating task or verifier issues. Distractor skills are the five most semantically similar SkillsBench community-corpus skills for each task domain.

Table 5: Agent-level selector comparison on WildClawBench (72 runs). Verifier scores on a 0–1 scale; bold: best.

Condition	Mean score	Success rate
Graph (fixed path-graph)	0.360	0.94
LTR (path-graph + LTR)	0.440	1.00
Agent all-skills (no selector)	0.438	0.94
Contract-aware (ours)	**0.488**	1.00

All conditions use identical agent, tasks, and verifiers. Contract-aware vs. Graph: +0.128, bootstrap 95% CI [0.016, 0.242], $p = 0.058$. Contract-aware vs. LTR: +0.047, CI [0.013, 0.085], $p = 0.029$. Contract-aware vs. Agent all-skills: +0.050, CI [−0.055, 0.154], $p = 0.81$.

shows a small penalty of 0.022—the community-authored calendar skills happen to share sufficient contract vocabulary for the verifier to award partial credit.

Agent-level selector comparison. The sensitivity experiments establish that misrouting causes harm; we further ask whether our contract-aware selector improves downstream task performance relative to alternative selectors. We conducted a 72-run experiment on WildClawBench (6 Social Interaction tasks, 4 conditions, 3 repeats per condition): (i) `graph_contract` (our method, selects one skill via contract-aware path-graph), (ii) `path_graph_ltr` (learning-to-rank on path-graph features), (iii) `graph` (fixed path-graph), and (iv) `all` (agent sees all six skills and selects implicitly). All conditions use the same agent, tasks, and verifiers; only the skill exposure mechanism differs.

Our method achieves the highest mean verifier score (0.488), significantly outperforming the graph baseline (+0.128) and the LTR selector (+0.047). The agent all-skills condition is competitive on average but inconsistent across tasks (e.g., agent wins on meeting-negotiation where exploring multiple workflows helps, but loses on action-extraction and multi-step reasoning where contract discrimination matters). This pattern is consistent with our case study (Section 5.7): agent self-selection sometimes succeeds but is unreliable—it can select a skill from the wrong family entirely. A dedicated contract-aware selector provides consistent, auditable routing decisions without relying on the agent to discover and evaluate skills at runtime.

Table 6: Evidence-gated correction on two base rankers (120 queries). Improves both while never harming them. Bold: corrected Top-1.

Base ranker	Base	Corrected	Δ	Invoked	Accepted
Path-graph	0.6917	**0.7417**	0.0500	101	12
Semantic text	0.7167	**0.7667**	0.0500	63	15

Together, the two sensitivity experiments provide complementary ecological validation on an independently published benchmark with real skills, real human-authored tasks, and automated verifier scores. Internal WB skill confusion shows penalties up to 0.360 when the agent receives a different WB-authored skill. External community-skill confusion (Table 4) shows penalties up to 0.514 when the agent receives SkillsBench community-corpus skills that share surface vocabulary but differ in operational contracts. Both experiments confirm the central claim: skill selection quality causally determines downstream task success, and the penalty magnitudes far exceed the offline ranking gaps reported in Section 5.2 (internal skill confusion: 0.093–0.360; external community-skill confusion: 0.022–0.514). These results demonstrate that WildClawBench contains a subset of tasks where operational-contract ambiguity is the decisive failure mode, and contract-aware routing provides a systematic improvement over existing selectors on this subset.

5.5 Evidence-Gated Correction (RQ3)

RQ3 asks whether evidence-gated correction improves base rankers without regression. Table 6 shows correction results on two base rankers. On path-graph, correction improves from 0.6917 to 0.7417 (+0.0500, recovering 6 of the 37 queries path-graph misses, a 16% error reduction), accepting 12 of 101 low-margin cases (95% bootstrap CI of the delta: [0.017, 0.092], $p < 0.05$). On semantic text, correction improves from 0.7167 to 0.7667 (+0.0500, recovering 6 of the 34 queries semantic misses), accepting 15 of 63 low-margin cases (95% CI: [0.000, 0.100], borderline). In all cases, correction never reduces accuracy below the base ranker.

The gate is intentionally conservative: it accepts only corrections where grounded operational-contract evidence is sufficient, falling back to the base ranker otherwise. On path-graph, 101 of 120 queries have low margin and the schema is invoked; 12 corrections are accepted (12%), of which 6 are correct (precision = 50%). The remaining 89 are rejected and fall back to path-graph, which was already correct on 64 of those 89 cases. The net effect among invoked cases is 64 correct (base ranker alone) → 70 correct (with correction + fallback), a gain of 6. On semantic text, 63 cases are invoked, 15 accepted (precision = 53%), yielding 32 → 38 correct. The key property is that rejected corrections cause *no harm*: the selector falls back to the base ranker's original decision.[1] Gate threshold sensitivity analysis (low, default, and high settings) leaves Top-1 accuracy unchanged at 0.7833 while the trigger rate varies from 100% to 75%, confirming that the correction layer is robust to threshold choice.

[1]Among the 12 accepted path-graph corrections, none reduced accuracy below the base ranker (0 observed regressions; 95% binomial CI on regression rate: [0%, 22.1%]). While the sample limits precision, the zero count is consistent with the gate's design—it selects among candidate skills rather than overriding the ranker arbitrarily.

5.6 Regime Boundary and Component Analysis (RQ4)

RQ4 asks whether the method is regime-specific. We evaluate on 30 queries over five families from the fixed SkillsBench community-corpus snapshot (code-review, frontend-design, brainstorming, git-workflow, testing), each with 6 real skill variants from different community authors on skillsmp.com [10]. These skills are pure text artifacts—coding guidelines, design patterns, workflow checklists—without the structured API endpoints and operational contracts that ContractBench skills encode. On this set, **all offline routing methods perform at random level**: mean Top-1 across structured and semantic strategies is 0.15 versus the random baseline of 0.17 (6-way classification). No offline method exceeds random by a statistically significant margin. Notably, even LLM-based selection with description-only access scores 0% on this subset (Section 5.3, Table 3), confirming that the challenge is fundamental—skills without operational contracts cannot be routed by any method, structured or neural.

Our method's random-level performance is not a weakness but a design property: when operational contracts are absent, the evidence gate finds no grounded contract fields and correctly defers to the base ranker. A deployed system can implement a lightweight regime detector—checking whether the query contains contract-relevant cues (source-of-truth language such as "live" or "snapshot"; audience terms such as "internal" or "customer"; evidence requirements such as "trace" or "reference"; and format specifications such as "JSON" or "markdown")—to decide whether to invoke schema induction or fall back to the base ranker. This avoids unnecessary computation for general-purpose skills while activating contract-aware routing for the subset of queries where it matters—which, as the fixed corpus snapshot statistics show (34K+ skills, 356 names with ≥6 variants), is a large and growing regime.

Component ablation. Table 7 reports ablations that remove key components. Removing individual contract dimensions (required capability, forbidden capability, side effect, postcondition, hard constraints) leaves both Top-1 accuracy and rank order unchanged—these fields encode complementary constraints captured by other evidence dimensions. Three ablations each reduce Top-1 by 0.008 (from 0.7833 to 0.7750): removing evidence-span validation, replacing field-level semantic matching with whole-document matching (`no_field_level`), and replacing query-conditioned schema induction with a blind schema (`no_schema`). The identical drop indicates they degrade the same underlying mechanism: evidence quality.

While the aggregate Top-1 shift appears small, its impact is concentrated in precisely the regime where the method provides value. Path-graph fails on 37 of 120 queries; query-rules corrects 11 of these. Each ablation changes 10 selections overall (8.3%), but **all 10 changes occur within the 37 path-graph-error queries** (27.0% of that subset), and **zero changes occur in the 83 queries where path-graph is already correct**. On the 11 queries that query-rules uniquely fixes over path-graph, each ablation breaks 4 of those fixes (36.4%)—the components are essential to the method's core value proposition. The harmful changes are concentrated in incident_ops (2 of 4) and security (2 of 4), the families where query-rules shows the largest gains over path-graph (Table 2). Conversely, the ablations never regress below path-graph's baseline: unlike query-rules' 78.3% Top-1, the ablated variants at 77.5% remain above path-graph's 69.2%, and no ablated selection reverts to path-graph's error on queries where path-graph was correct.

Table 7: Component ablation (120 queries). Bold: best Top-1.

Variant	Top-1	ΔSel.	↓Acc.	PG-err%	QR-fix%
Full query-rules	**0.7833**	—	—	—	—
− evidence-span	0.7750	10	4	27.0%	36.4%
− field-level	0.7750	10	4	27.0%	36.4%
− schema	0.7750	10	4	27.0%	36.4%

ΔSel.: number of selections changed. ↓Acc.: number where accuracy decreased. PG-err%: fraction of path-graph's 37 errors affected. QR-fix%: fraction of query-rules' 11 unique fixes broken. Contract-dimension ablations (not shown) leave Top-1 unchanged.

As a complementary check, we evaluate a `graph_conventional` variant on the Social Interaction benchmark (6 WB tasks) that strips all structured contract dimensions—endpoints, actions, safety modes, and penalty weights—from the path-graph scorer, leaving only text similarity. Top-1 accuracy drops from 1.00 to 0.50. This confirms that structured contract dimensions, not text similarity, drive discrimination on tasks where endpoints and actions encode decisive contract differences.

5.7 Case Studies

Why field-level decomposition succeeds where whole-doc methods fail. We examine two contrasting cases from ContractBench, one where query-rules uniquely succeeds and one where it fails, to illustrate the method's decision logic.

Case A (query-rules win). Task 001 (incident_ops, live_internal_json). The query specifies four contract dimensions in abstract language: "Base the analysis on the current system state" (live data source), "output is for an internal team" (internal audience), "trace each finding back to its origin with a reference" (evidence traceability), and "produce a structured machine-readable record" (JSON output). Query-rules extracts these four fields with evidence spans anchored in the query text, then matches them against each candidate skill's body using field-level ST similarity. The live_internal_json skill body references "real time," "internal operations," "machine-readable JSON," and "message ID"—all scoring above threshold on the corresponding contract dimensions. Semantic BOW (family-level scores: 0.167 for this query) and whole-doc ST (0.083) both fail here because the shared Slack/Gmail/calendar vocabulary across all six candidates dilutes the contract-level signal. Path-graph also fails (0.167, no contract dimensions in its ontology). Query-rules (1.000) is the only method that correctly selects the live_internal_json skill.

Case B (query-rules lose). Task 038 (finance_reconcile, snapshot_engineering_md). The query says "Use the static record of the workspace state" (snapshot), "audience is a technical review team" (engineering), and "create a review document" (markdown). However, the query also mentions "precise references" and "relevant section of the source material" in its traceability clause. The schema induction module extracts an evidence requirement that over-indexes on "references" and "source material," producing a field description semantically closer to "search index" than "document span." The field-level matcher then assigns higher scores to

the search_finance_csv skill (whose body emphasizes "document search index" and "spreadsheet cell" references). Semantic BOW succeeds here (1.000 for this query) because its whole-document vocabulary overlap captures the "snapshot" and "engineering" terms directly in the skill body without the distortion introduced by field-level decomposition. Query-rules (0.000) fails. This case illustrates a failure mode noted in the per-family analysis: families with policies whose evidence traces share vocabulary with other contract dimensions are vulnerable to schema induction drift.

Agent self-selection failure on ContractBench. We also ran an agent self-selection experiment on ContractBench task 001 (incident_ops, live_internal_json policy). The agent was given all six candidate skills and instructed to route the request. The agent explored the workspace directory, discovered unrelated skill descriptions from legacy WildClawBench tasks, and selected `03_task5` (Support Triage)—a skill from a completely different domain. The agent's transcript shows confident but incorrect reasoning. Meanwhile, our evidence-gated correction selector correctly identified `incident_ops_live_internal_json` (the correct ContractBench skill) after correcting path-graph's initial choice. The agent chose a skill from the wrong skill family; the selector chose the correct skill within the right family. While anecdotal ($n = 1$), this case illustrates why routing benefits from a separate control layer: even with full filesystem access to candidate skills, the agent may fail to discover, read, and correctly evaluate them.

6 Conclusion

The operational-contract under-specification problem arises when candidate skills share coarse execution paths but differ in source-of-truth, recipient scope, evidence policy, and output format. Existing routing methods—fixed structured ontologies and semantic text embeddings—fail in this regime because they conflate domain vocabulary (shared) with contract vocabulary (discriminative). We proposed query-conditioned operational schema induction with auditable, evidence-gated correction: a declarative `QueryRuleSpec` is induced from the query alone, deterministic extraction produces evidence-bearing `OperationalSignature` objects, and an evidence gate accepts correction only when grounded contract fields are present, falling back to the base ranker otherwise.

Experiments across ContractBench and WildClawBench support four conclusions. First, skill selection quality causally determines downstream task success: across seven WildClawBench tasks, misrouting reduces verifier scores by a mean of 0.187–0.228 on a 0–1 scale, with individual penalties reaching 0.514. Second, field-level decomposition bridges the gap that whole-document methods miss: `query_rules` reaches 0.7833 Top-1 versus 0.6917 for path-graph and 0.7167 for semantic text. Third, evidence-gated correction improves both tested base rankers (recovering 6 of 37 path-graph errors and 6 of 34 semantic errors) with a zero-regression property—empirically never reducing accuracy below the base ranker. Fourth, the method is regime-aware: on the SkillsBench community-corpus subset without operational contracts, the evidence gate defers to the base ranker and all methods perform at random level.

Our method has several limitations. ContractBench is targeted and does not provide prevalence estimates for deployed libraries. WildClawBench task exclusions were determined post-hoc. The evidence gate is conservative and depends on the query exposing enough operational information. path-graph should remain the default when its fixed ontology is confident. The method is designed for the operational-contract regime and does not claim to improve routing for general text skills. Two directions merit further investigation: extending schema induction with LLM-based extraction for broader coverage, and closing the selector-to-execution loop.

GenAI Usage Disclosure

The authors used generative AI tools, including OpenAI ChatGPT/Codex, to assist with software implementation, experiment orchestration, result summarization, and drafting/editing of this manuscript. All code changes, experiment commands, generated tables, and claims were inspected by the authors against local run records and verifier outputs. Generative AI tools were not used to fabricate experimental data or alter verifier results.

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